

MOSTOFA HASIN MAHDI

hasinmahdi.hmr@gmail.com · +8801799407741 · Uttara, Dhaka, Bangladesh · [GitHub](#) · [LinkedIn](#) · [Portfolio](#)

SUMMARY

Final-year Computer Science and Engineering student with hands-on experience building backend APIs, data-driven applications, and AI/ML systems using Python, FastAPI, React, PostgreSQL, and Scikit-learn. Experienced in developing OCR pipelines, multi-tenant systems, API validation workflows, and machine-learning applications.

EDUCATION

Shanto-Mariam University of Creative Technology

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

2022 – Expected Nov 2026

CGPA: 3.96

PROJECTS

NAB Preg AI

FastAPI, Next.js, XGBoost, Scikit-learn, Mistral AI, Supabase

Lead ML Engineer (Team of 5)

- Built a multi-output ML model predicting 3 clinical risk targets in **<50ms** with **90% accuracy**.
- Mitigated class imbalance by augmenting **740 records with 400 synthetic high-risk profiles** using WHO-aligned thresholds.
- Developed an **OCR pipeline** integrating Mistral's **Pixtral-12B** via FastAPI, parsing prescriptions and lab reports into sanitized JSON for frontend validation.
- Engineered a **5-step OCR failover** pipeline using 4 backup Mistral API keys and Tesseract.js as a local fallback.
- Implemented dual-layer Zod/Pydantic validation to sanitize 10 patient fields and prevent malformed database payloads.
- *Demo:* [Live Demo](#) · [GitHub](#)

PDF to Markdown Conversion Engine

PyMuPDF, pdfplumber, Pandas, spaCy, PyTesseract

Solo Developer

- Developed an open-source Python CLI using custom **coordinate-based** layout detection to sort text, detect headings, and prevent overlapping content.
- Integrated **spaCy-based text cleaning** to resolve sentence fragmentation and PyTesseract OCR for scanned pages.
- Engineered table extraction that identifies complex grids and reconstructs them into aligned Markdown tables.
- *Demo:* [GitHub](#) · [Release](#)

Bizit Business Suite

React, FastAPI, Pydantic, PostgreSQL, Pytest

Solo Developer

- Engineered a secure multi-tenant architecture using PostgreSQL data scoping, ensuring **zero cross-tenant leaks**.
- Implemented 3-tier RBAC (Owner/Admin/Employee) and Pydantic validation across **24 API endpoints**.
- Optimized PostgreSQL queries to aggregate **10,000+ sales and cost records** with **sub-10ms** response times.
- Wrote **17 Pytest integration tests** covering Stock, Sales, Suppliers, RBAC, and tenant isolation, executed automatically on every push via GitHub Actions CI.
- *Demo:* username: hasin, password: hasin123 · [Live Demo](#) · [GitHub](#)

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL
Backend/Frontend: FastAPI, Next.js, React, REST APIs, Pydantic
AI/ML: Scikit-learn, XGBoost, Pandas, OCR, NLP
Databases: PostgreSQL, MongoDB, MySQL
DevOps & Tools: Docker, Git, GitHub Actions, Pytest, Linux

ACHIEVEMENTS

- **Champion**, SMUCT Software Project Showcase 2026; **Finalist**, CloudcampBD Infinity AI BuildFest 2026.
- **2nd Runner-up**, SMUCT CSE FEST V2.0 Programming Contest; participated in BUET & SUST IUPC Prelims and ICPC Regional Prelims (2022–2023).